Z-Scores & Linear Transformations

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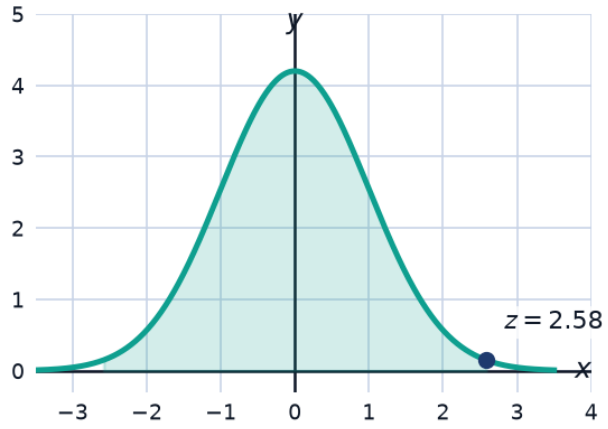
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Computing z-scores**

A test has mean 70 and standard deviation 5. Find the z-score for a student who scored 82.

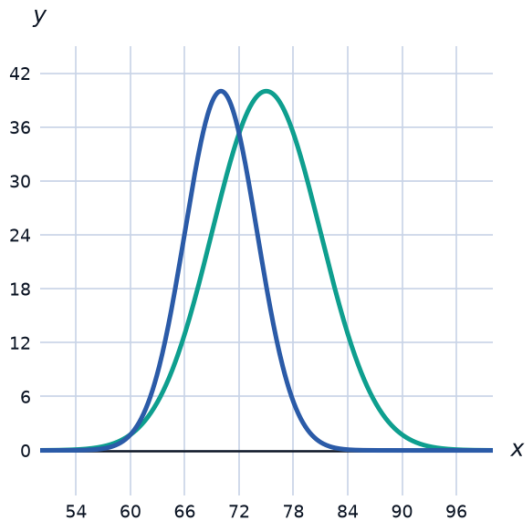
2 **[Foundational]** • **Target: Computing z-scores**

Explain in words what a z-score of -1.5 means, in the context of a data value's position relative to its distribution.



3 **[Foundational]** • **Target: Comparing values across different distributions**

On Test A (mean 75, standard deviation 6), a student scores 87. On Test B (mean 70, standard deviation 4), a different student scores 77. Compare the two students' relative performance using z-scores.



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4 [Foundational] • Target: Effect of adding a constant to every data value

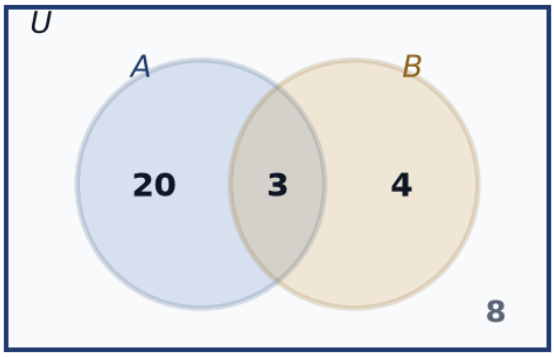
A data set has mean 50 and standard deviation 8. If 10 is added to every value, find the new mean and new standard deviation.

5 [Foundational] • Target: Effect of adding a constant to every data value

Explain, without doing a full calculation, why adding a constant to every data value does not change the standard deviation.

6 [Foundational] • Target: Effect of multiplying every data value by a constant

A data set has mean 20 and standard deviation 3. If every value is multiplied by 4, find the new mean and new standard deviation.



7 [Foundational] • Target: Effect of multiplying every data value by a constant

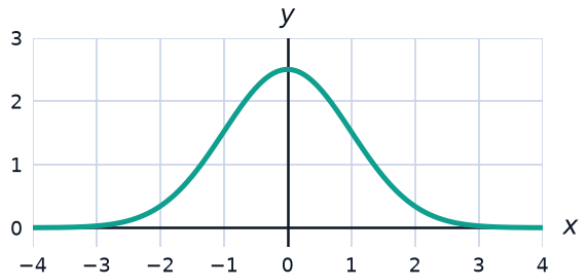
A distance measured in miles has mean 10 and standard deviation 2. Convert both to kilometers (multiply by 1.609), and find the new mean and standard deviation, to two decimal places.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Visualizing a standardized (z-score) distribution

The graph shows the standard normal distribution, which has mean 0 and standard deviation 1 — the distribution that results after standardizing any normal data set using z-scores.



a State the mean and standard deviation shown by this curve.

b Explain why standardizing a data set using $z = \frac{x - \mu}{\sigma}$ always produces a distribution with mean 0 and standard deviation 1, regardless of the original data's mean and standard deviation.

9 [Intermediate] • Target: Applying linear transformations in context

A factory's daily output (in units) has mean 500 and standard deviation 30. Management adds a fixed bonus batch of 50 units to every day's count, then reports output in dozens (divide by 12). Find the new mean and standard deviation, to two decimal places.

10 [Intermediate] • Target: Finding a raw value from a z-score

A distribution has mean $\mu = 64$ and standard deviation $\sigma = 5$. Find the data value corresponding to a z-score of $z = -1.8$.

11 [Intermediate] • Target: Finding a raw value from a z-score

A distribution has mean $\mu = 200$ and standard deviation $\sigma = 15$. If a data value has $z = 2.2$, find the data value.

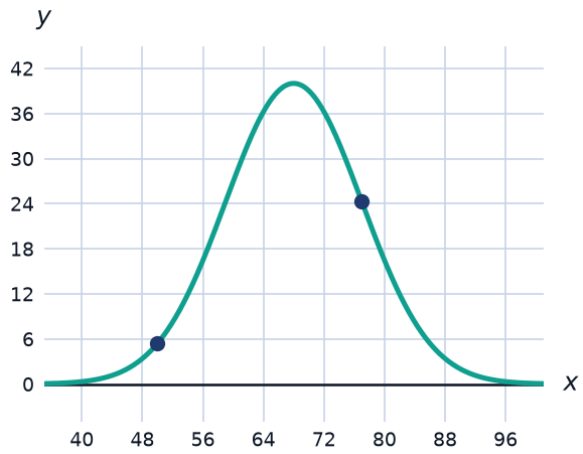
12 [Intermediate] • Target: Combined shift-and-scale transformations

A teacher curves test scores (mean 65, standard deviation 10) by first adding 5 points to every score, then multiplying the result by 1.1. Find the new mean and new standard deviation, to two decimal places.

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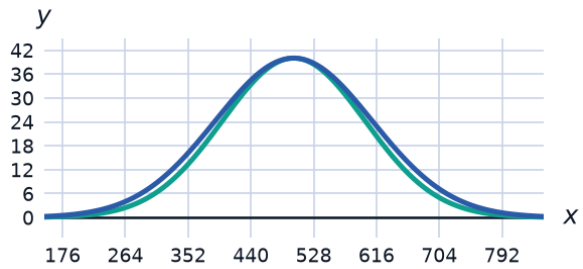
13 [Intermediate] • Target: Combined shift-and-scale transformations

A weight distribution has mean $\mu = 68$ kg and standard deviation $\sigma = 9$ kg. Find the z-scores for weights of 77 kg and 50 kg.



14 [Intermediate] • Target: Combined shift-and-scale transformations

A national math exam has mean 500 and standard deviation 100, while a national verbal exam has mean 500 and standard deviation 110. A student scores 640 on the math exam and 625 on the verbal exam. Even though the verbal raw score is lower, use z-scores to determine which exam the student performed better on relative to other test-takers, and explain your reasoning.



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Combined shift-and-scale transformations • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

A shipping company records package weights with mean 18 kg and standard deviation 2.5 kg. The company then converts every measurement from kilograms to pounds by multiplying by 2.20462, and separately subtracts 1 kg from every package to account for packaging weight before converting. Describe, step by step, how each operation (the subtraction, then the multiplication) affects the mean and standard deviation.

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16 [Challenging] • Target: Combined shift-and-scale transformations • Cross-Lesson: 3.03 The Normal Distribution

Level 3 · AdvancedA scatterplot displays a point at $x = 18$, where the rest of the sample data has x-values between 2 and 8. The point falls almost exactly along the linear trend established by the other points. Explain the effect of this point on: (a) leverage, (b) the residual, (c) the correlation coefficient r, and (d) the slope b_1 of the LSRL.

17 [Challenging] • Target: Computing z-scores • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

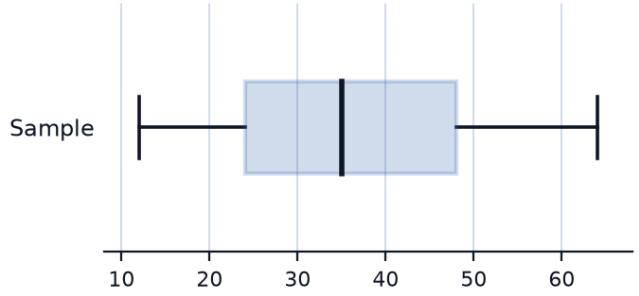
A least-squares regression line predicting exam score (y) from homework completion percentage (x) is $\hat{y} = 38.0 + 0.55x$ with correlation $r = 0.82$. (a) Interpret the slope 0.55 in context. (b) Calculate and interpret the coefficient of determination r^2.

18 [Challenging] • Target: Comparing values across different distributions • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

Level 1 · FoundationsAn airline wants to inspect passenger satisfaction. It randomly selects 10 flights departing from a major hub on a given day and surveys every passenger on those 10 flights. Identify this sampling method and distinguish it from stratified sampling.

19 [Challenging] • Target: Effect of adding a constant to every data value • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

A sample dataset has $Q_1 = 48$ and $Q_3 = 76$. Using the 1.5 IQR rule: (a) Determine the exact lower and upper outlier fences. (b) If the dataset includes values of 4 and 122, determine whether each is an outlier.



20 [Challenging] • Target: Effect of multiplying every data value by a constant • Cross-Lesson: 1.01 Measures of Center & Spread

In an SRS of 200 registered voters, 116 favor a new environmental regulation. (a) Check all three conditions for constructing a 95% confidence interval for the population proportion. (b) Construct the 95% confidence interval. (c) Interpret the interval in context.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Computing z-scores

A distribution has mean $\mu = 51$ and standard deviation $\sigma = 25$. Find the z-score for a value of $x = 41$.

a. -0.40

b. 0.40

c. -0.90

d. 0.10

22 [Foundational] • Target: Comparing values across different distributions

On Test A (mean 79, standard deviation 4), a student scores 94. On Test B (mean 88, standard deviation 8), a different student scores 99. Using z-scores, which student performed better relative to their own test's distribution?

a. Test A

b. Test B

c. Cannot be determined without more information

d. Both are exactly the same

23 [Foundational] • Target: Effect of adding a constant to every data value

A data set has mean 75 and standard deviation 17. First, -4 is added to every value; then every value is multiplied by 4. Find the new mean and standard deviation as an ordered pair (mean, SD).

a. $(284, 17)$

b. $(285, 68)$

c. $(284, 68)$

d. $(300, 68)$

24 [Foundational] • Target: Effect of multiplying every data value by a constant

A distribution has mean $\mu = 97$ and standard deviation $\sigma = 4$. Find the z-scores for values of 113 and 84, as an ordered pair (z_{113}, z_{84}), to two decimal places.

a. $(4.30, -3.25)$

b. $(4.00, -3.25)$

c. $(4.00, -3.55)$

d. $(-3.25, 4.00)$

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25 [Foundational] • Target: Visualizing a standardized (z-score) distribution

A team of statisticians is designing a complex experiment to evaluate the efficacy of a new fertilizer on crop yield across multiple regions and soil types. They have identified 3 distinct geographic regions and 5 different soil types. Within each region-soil type combination, they plan to test 4 different concentrations of the fertilizer. To account for variability, they will run 2 independent replicates of each treatment combination. However, due to resource limitations, they can only afford to run 150 unique treatment combinations in total. The team decides to allocate the available resources by prioritizing the combinations that are most likely to reveal significant differences, while ensuring a balanced representation across the major factors. They will select a subset of the possible treatment combinations. If they decide to use a stratified sampling approach where each region is a stratum, and within each region, they stratify by soil type, and then sample fertilizer concentrations, what is the minimum number of unique treatment combinations (region, soil type, concentration) they *must* include in their sample to ensure that every soil type is represented in at least one region's stratum, and every fertilizer concentration is represented within each soil type stratum across all regions, given that they still wish to have 2 replicates per selected combination?

a. 150

b. 90

c. 75

d. 60

26 [Foundational] • Target: Applying linear transformations in context

A discrete random variable X can take on the values 1, 2, 3, and 4. The probability distribution for X is given by P(X=x) = kx, where k is a constant. What is the value of P(X=2)?

a. $\frac{1}{6}$

b. $\frac{1}{3}$

c. $\frac{1}{2}$

d. $\frac{5}{6}$

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27 [Foundational] • Target: Finding a raw value from a z-score

A biostatistician is investigating the relationship between the dosage of a new experimental drug (measured in milligrams) and the reduction in systolic blood pressure (in mmHg) for patients with mild hypertension. They collected data from $n = 200$ patients and fit a simple linear regression model. The model's output yielded an estimated intercept of $\hat{\beta}_0 = 5.2$ and an estimated slope of $\hat{\beta}_1 = -0.8$. The standard error of the estimated slope is $\text{SE}(\hat{\beta}_1) = 0.15$. A 95% confidence interval for the true slope β_1 was calculated to be $[-1.12, -0.48]$. The biostatistician then constructs a hypothesis test to determine if the true slope is significantly different from zero. To do this, they calculate a test statistic. If the same α level used for the confidence interval were applied to a two-tailed hypothesis test of $H_0 : \beta_1 = 0$ versus $H_a : \beta_1 eq 0$, what would be the approximate p-value associated with this test statistic?

a. Greater than 0.05

b. Equal to 0.05

c. Less than 0.05

d. Cannot be determined from the given information

28 [Foundational] • Target: Combined shift-and-scale transformations

A manufacturing company produces electronic components. They are investigating potential relationships between the type of raw material used and the number of defects per batch. After collecting data for 100 batches, they create a two-way table summarizing the findings. The table shows that 15 batches using Material A had no defects, 20 batches using Material A had 1-2 defects, and 10 batches using Material A had more than 2 defects. For Material B, 25 batches had no defects, 28 batches had 1-2 defects, and 22 batches had more than 2 defects. If the company decides to perform a chi-square test for homogeneity to compare the defect rates across the two materials, what is the degrees of freedom for this test?

a. 2

b. 3

c. 4

d. 5

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 **[Intermediate]** • **Target: Computing z-scores**

A research team is investigating the relationship between socioeconomic status (SES) and preferred mode of transportation for commuting to work in a large metropolitan area. They conduct a survey and categorize SES into three levels: Low, Medium, and High. The preferred mode of transportation is categorized as Public Transit, Private Vehicle, or Bicycle/Walking. After collecting data from a sample of $n = 500$ individuals, they decide to perform a chi-square test for independence. The preliminary analysis suggests that the expected counts for several cells are small. To ensure the validity of the chi-square approximation, the research team decides to merge categories. They consolidate 'Bicycle/Walking' into 'Public Transit' to form a new category 'Active/Public Transit'. If the original analysis involved 3 levels of SES and 3 modes of transportation, what is the minimum number of observations required in each cell of the *original* contingency table for the chi-square test to be considered reasonably reliable, assuming equal distribution across all cells?

a. 4

b. 5

c. 6

d. 10

30 **[Intermediate]** • **Target: Comparing values across different distributions**

A local government is analyzing the relationship between the socioeconomic status (SES) of residents in a city and their reported level of civic engagement. They surveyed a random sample of 500 residents and categorized them into three SES levels: Low, Medium, and High. Civic engagement was also categorized into three levels: None, Some, and High. The results are summarized in the two-way table below. A chi-square test for homogeneity is planned to investigate if the distribution of civic engagement levels differs across SES groups. | SES Level | None | Some | High | Total | |---|---|---|---|---| | Low | 60 | 90 | 50 | 200 | | Medium | 70 | 80 | 50 | 200 | | High | 30 | 40 | 30 | 100 | | Total | 160 | 210 | 130 | 500 | If the null hypothesis of homogeneity is rejected, which of the following statements best describes a potential conclusion from this test, assuming the observed counts are representative of the city's population?

a. There is no association between socioeconomic status and civic engagement.

b. Socioeconomic status has no impact on the distribution of civic engagement levels.

c. The proportion of residents with high civic engagement is the same across all socioeconomic status levels.

d. The distribution of civic engagement levels significantly differs across the socioeconomic status groups.

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31 [Intermediate] • Target: Effect of adding a constant to every data value

A researcher is analyzing the relationship between the number of hours a student spends on extracurricular activities per week and their final exam score in a challenging STEM course. The researcher collects data from 100 students and fits a linear regression model. The model is given by $\text{Exam Score} = 65 + 2.5 \times \text{Hours} - 0.3 \times \text{Hours}^2$, where Exam Score is the final exam score and Hours is the number of hours spent on extracurricular activities per week. The researcher believes that for a certain range of hours, there might be an optimal number of extracurricular hours that maximizes the exam score, and beyond that, the score might decrease due to time constraints. \\ \\ Consider the function representing the exam score as a function of hours: $f(h) = 65 + 2.5h - 0.3h^2$. To find the maximum exam score and the corresponding hours, calculus can be used. \\ Which of the following is the approximate number of extracurricular hours per week that would maximize the student's final exam score, assuming the model holds for $h \geq 0$?

a. 4.17 hours

b. 2.5 hours

c. 5.0 hours

d. 8.33 hours

32 [Intermediate] • Target: Effect of multiplying every data value by a constant

A political scientist is analyzing voter preferences for two candidates, Candidate A and Candidate B, across different age groups. The data is summarized in the two-way table below. The total number of surveyed individuals is $N = 500$. The total number of individuals in the 'Young Adults' row is $R_1 = 120$, and the total number of individuals who prefer Candidate A (Column 1) is $C_1 = 200$. If the political scientist were to conduct a chi-square test of independence on this data, what would be the expected count for the cell representing 'Young Adults' who prefer 'Candidate A'?

a. 48

b. 52

c. 50

d. 72

33 [Intermediate] • Target: Visualizing a standardized (z-score) distribution

The scatterplot below displays the relationship between the number of hours (h) students spent studying for a math test and their scores (s) on that test. The equation of the least-squares regression line for this data is given by $s = 1.2h + 78.5$. If a student studied for 5 hours, what is their predicted score based on this model?

a. 91.5

b. 84.5

c. 81.5

d. 90.5

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34 [Intermediate] • Target: Applying linear transformations in context

The scatterplot below displays the relationship between the number of hours students spent studying for a test and their scores on that test. The equation of the least-squares regression line is approximately $\hat{y} = 55.2 + 4.8x$, where x is the number of hours spent studying and $\hat{y}$ is the predicted test score.

a. 55.2

b. 4.8

c. 50

d. 70

35 [Intermediate] • Target: Finding a raw value from a z-score

A researcher is studying the relationship between study hours and final exam scores for a group of AP Statistics students. The researcher collected data and constructed a scatterplot to visualize the relationship. The scatterplot includes a line of best fit. The equation of the line of best fit is given by $\hat{y} = 2.5x + 15$, where x represents the number of hours studied per week, and $\hat{y}$ represents the predicted final exam score. If a student studied for 12 hours, what is the predicted final exam score for that student?

a. 99

b. 45

c. 30

d. 60

36 [Intermediate] • Target: Combined shift-and-scale transformations

The following diagram shows a scatterplot comparing the number of hours studied per week for a math course and the final exam score for a sample of students. What is the approximate y-intercept of the line of best fit?

a. 50

b. 60

c. 70

d. 80

37 [Intermediate] • Target: Computing z-scores

A distribution is displayed as a histogram with a long tail extending to the right, most values clustered on the left. How should this shape be described?

a. Skewed right (positively skewed)

b. Bimodal

c. Skewed left (negatively skewed)

d. Approximately symmetric (unimodal)

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Comparing values across different distributions • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

A distribution is displayed as a histogram with a long tail extending to the left, most values clustered on the right. How should this shape be described?

a. Skewed left (negatively skewed)

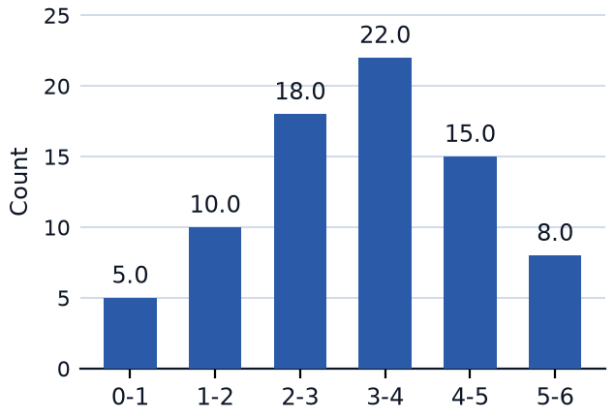
b. Approximately symmetric (unimodal)

c. Skewed right (positively skewed)

d. Bimodal

39 [Challenging] • Target: Effect of adding a constant to every data value • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

The histogram below displays the distribution of the number of hours students spent studying for a recent exam. What is the approximate median number of hours studied?



a. 2.5 hours

b. 3.5 hours

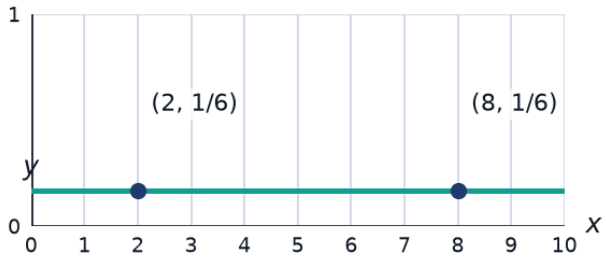
c. 4.5 hours

d. 5.5 hours

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40 **[Challenging]** • **Target: Effect of multiplying every data value by a constant** • **Cross-Lesson: 1.01 Measures of Center & Spread**

Consider a continuous random variable X with a uniform distribution over the interval $[2, 8]$. What is the probability that X is between 4 and 7, i.e., $P(4 < X < 7)$?



a. 0.50

b. 0.25

c. 0.75

d. 0.17

41 **[Challenging]** • **Target: Visualizing a standardized (z-score) distribution** • **Cross-Lesson: 1.03 Describing Distributions & Linear Regression**

Consider a discrete random variable X that represents the number of heads obtained when a fair coin is tossed 3 times. The probability distribution of X is given by: | x | 0 | 1 | 2 | 3 | |---------|-------|-------|-------|-------| | $P(X = x)$ | $\frac{1}{8}$ | $\frac{3}{8}$ | $\frac{3}{8}$ | $\frac{1}{8}$ | What is the probability that X is less than 2, i.e., $P(X < 2)$?

a. $\frac{3}{8}$

b. $\frac{4}{8}$

c. $\frac{5}{8}$

d. $\frac{6}{8}$

42 **[Challenging]** • **Target: Applying linear transformations in context** • **Cross-Lesson: 2.01 Sampling Methods & Bias**

A town council is conducting a survey about a proposed new park. They decide to use a stratified random sample to ensure representation from different neighborhoods. The total population of the town is 45,000. The town is divided into three neighborhoods: Northside (18,000 residents), Eastside (15,000 residents), and Westside (12,000 residents). The council plans to survey a total of 900 residents. If the sample is to be proportional to the population of each neighborhood, how many residents should be surveyed from the Northside neighborhood?

a. 360

b. 450

c. 300

d. 400

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43 [Challenging] • Target: Finding a raw value from a z-score • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

A researcher is designing a complex experiment to investigate the impact of three different fertilizer treatments (A, B, C) on crop yield, while also controlling for two different irrigation methods (I1, I2) and two soil types (S1, S2). To ensure a robust comparison and account for potential interactions, a stratified randomized block design is employed. Within each unique combination of irrigation method and soil type (forming the blocks), the three fertilizer treatments will be randomly assigned to distinct plots. What is the total number of unique experimental units (plots) that will be established for this experiment?

a. 12

b. 6

c. 3

d. 18

44 [Challenging] • Target: Combined shift-and-scale transformations • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

An agricultural researcher is designing an experiment to study the yield of a new variety of wheat. The experiment investigates two factors: - Factor 1: Fertilizer formula, with 3 different types. - Factor 2: Watering frequency, with 4 different schedules. If the experiment uses a completely randomized design where each subject receives a combination of one fertilizer formula and one watering schedule, how many distinct treatment groups are there?

a. 7

b. 12

c. 24

d. 64

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45 [Challenging] • Target: Computing z-scores • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

A large university is conducting a comprehensive study on student satisfaction with campus dining services. They decide to employ a complex sampling strategy involving multiple stages to ensure representation across various student demographics and living situations. The university has identified three main housing categories: On-Campus Dormitories (CD), Off-Campus Apartments (OA), and Commuters (CM). Within the On-Campus Dormitories, there are two distinct dining halls: North Campus Dining (ND) and South Campus Dining (SD). The total student population is 25,000. The university wants to select a sample that reflects the proportion of students in each housing category. Additionally, for students in On-Campus Dormitories, they want to ensure proportional representation from each dining hall. The initial allocation of the total sample size of 1000 students is as follows: 60% to On-Campus Dormitories, 30% to Off-Campus Apartments, and 10% to Commuters. For the On-Campus Dormitory sample, 70% are allocated to North Campus Dining and 30% to South Campus Dining. If the university wishes to further stratify the sample within Off-Campus Apartments based on student major (Humanities, STEM, Business), and they have determined that the proportion of Humanities majors in the Off-Campus Apartments is 0.40, STEM is 0.35, and Business is 0.25, what is the expected number of students from Off-Campus Apartments who are Humanities majors in the overall sample of 1000 students?

a. 100

b. 120

c. 140

d. 160

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